

FARZAN TOOR

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Technical Skills

CAD & Software: AutoCAD, C++, Excel, Minitab, SolidWorks (CSWA)

Fabrication & Manufacturing: Automotive painting, Carbon fiber fabrication, Metal fabrication, Autobody repair, MIG welding, GD&T, Machining

Experience

Ontario Drive & Gear

May 2026 – Aug 2026

Quality Engineer

Waterloo, ON

- Led in-depth capability analysis on 4 unique parts to determine corrective actions and improve part quality using Minitab and Excel.
- Generated 40+ inspection specifications with GD&T symbols from mechanical drawings, allowing for a standardized inspection process and assisting with production release readiness.
- Supported PPAP/FA submissions by generating capability presentations that revealed critical insights and guided key management decisions.
- Logged 200+ Non-Conformance Reports, contributing to enhanced quality control, participating in daily meetings, and managing the online NCR database.
- Developed extensive knowledge of gear manufacturing processes and quality inspection methods.

FSAE University of Waterloo Aerodynamics Team

Sept 2025 – Present

Aerodynamics Team Member

Waterloo, ON

- Designed whiskers for the front nosecone using SolidWorks for the 2027 car.
- Fabricated aero components using carbon fiber, and machined components such as bearings for the car.

Advantage Ford

Aug 2023 – Aug 2025

Shop-Hand Technician

Calgary, AB

- Performed fundamental automotive maintenance, including oil changes, tire balancing, and battery diagnostics, to develop a strong foundational understanding of vehicle mechanics.
- Analyzed complex engine failure cases alongside technicians—diagnosing issues such as coolant-oil mixing, excessive oil consumption, and engine block damage—to build practical knowledge of powertrain failure modes.
- Executed detailed quality inspections on factory-new Ford vehicles to identify transit defects, accurately documenting and reporting paint and wheel damage to Ford Canada.
- Safely operated and staged 20–35 vehicles per shift, successfully maneuvering a diverse fleet that included consumer cars, commercial vans, and heavy-duty trucks.

Projects

Custom 2WD Short Course RC Car

Mechatronics Design Project

- Engineering a high-speed, custom 3D-printed 2WD short course RC car, utilizing precision calipers to reverse-engineer existing components and modify them.
- Redesigning critical components of the RC car, such as the chassis, suspension system, and turning system, using SolidWorks.

Slash Gearbox Housing Redesign

Mechanical Redesign Project

- Reverse-engineering 7 internal gears utilizing a profile projector and specialized gear design software to extract critical geometry, including pitch diameter, module, and pressure angle.
- Modeling the complete gear train assembly in SolidWorks to design and tolerance an optimized, custom gearbox housing.

Education

University of Waterloo

Sept 2025 – Expected 2030

B.A.Sc., Mechatronics Engineering

Waterloo, ON

- President's Scholarship Recipient.